

DEV SPRINT 4 · STATUS REPORT

Agents Go Live.

Seventeen pull requests, thirty-eight issues, one continuous session. The full Model Registry & AI Gateway, a resolver-aware agent runtime, a Conductor that decomposes briefs and dispatches to seven role-agents, a workflow runner with five production templates, AEO scoring, observability + tech-debt — shipped, merged, tagged as **v1.1.2**.

SPRINT WINDOW

14 May – 15 May 2026

REPORT DATE

15 May 2026, 00:49 IST

PLAN COMPLETION

≈ 98 % of PLAN-v1.2
scope**Sprint 4 · DELIVERED · v1.1.2 TAGGED****Prepared by**

Engineering, with Claude Code · Opus 4.7

Confidential — internal distribution

v4.0 · 15.05.2026

IMPRINT

About this report

This is the official engineering and product status report for DClaw Marketing's fourth development sprint — the sprint that took the platform from a hardened, operator-ready v1.1.1 stack to an agent-driven runtime running real models, real role-agents, and real workflows under v1.1.2. It is intended for three audiences: **operators** (TL;DR), **executives** (Executive Summary), and **engineers** (Theme-by-Theme Status and onward).

FIELD	VALUE
Document title	DClaw Marketing — Dev Sprint 4 Status Report
Sprint number	Sprint 4 (open-ended cadence; Sprint 3 closed 2026-05-14 with the v1.1.1 release)
Sprint window	2026-05-14 23:28 IST → 2026-05-15 00:49 IST (≈ 1 h 21 m wall-clock; one intensive burst)
Sprint kick-off PR	#283 · docs(plan): Sprint 4 plan — Model Registry, AEO, competitive context
First dev PR	#322 · feat(models): S4-M1/M2 — Model Registry foundation (DB + CRUD)
Sprint closing PR	#339 · chore(release): v1.1.2 — Sprint 4 closeout (agents go live)
Snapshot ref	main @ HEAD · tag v1.1.2
PRs merged in window	17 (16 dev + 1 hotfix #338 + 1 release #339); plus the planning artefact #283
Issues closed	38 of 38 open dev issues (100 %); marketing #49–#53 off-limits to engineering
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How to read this document. If you have 5 minutes, read Chapter 1 (TL;DR). If you have 30 minutes, add Chapters 2–3 (Recap of Sprints 1 + 2 + 3 + Intent). If you are an engineer reviewing the work, read

everything from Chapter 4 onward; appendices contain the raw catalogue of every pull request shipped in the sprint.

What's new about this report

Chapter 2 carries a **combined recap of Sprints 1, 2 and 3**, so a reader returning to the project after a break has everything they need to make sense of Sprint 4's "agents go live" milestone. Sprint 1 laid the foundations; Sprint 2 took us to feature-complete; Sprint 3 hardened the operator surface; Sprint 4 wired the real agent runtime on top.

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CHAPTER 1

TL;DR

Sprint 4 was an **integration sprint**. v1.1.1 had hardened scaffolding and a clean operator surface but every agent call still used inline LLM stubs. Sprint 4 wired the real agent runtime on top: a Model Registry, a resolver, a Conductor that decomposes briefs, seven role-agents dispatched through a trust-mode policy, and a workflow runner that actually runs five production templates end-to-end. By closeout, an operator with valid API keys can drive the platform from a single chat box.

PRS MERGED

17

single ~1.4 h session

ISSUES CLOSED

38

100 % of dev scope

PLAN COMPLETION

≈ 98 %

PLAN-v1.2 + agent layer

OPEN ISSUES

5all marketing collaterals
· off-limits to engineering

What landed

- **Model Registry & AI Gateway (S4-M, 11 issues)** — 5 new DB tables, 4 Alembic migrations, 26 provider types across four tiers (Native APIs / Aggregators / Multimedia Specialists / Self-hosted), auto-discovery, capability heuristic with operator-override preservation, 5-min health-check Celery beat, ModelCallLog + SSE log stream + p50/p95/p99 metrics, Org & User assignment tables, resolver chain (user → org → pool → env → stub), /admin/models page with the 5-radio + Others ▾ provider picker, gate hook, inline action selectors.
- **Agent Runtime (S4-A, 5 issues)** — resolver-aware run_completion() over Anthropic + OpenAI-compatible providers, Conductor decompose_brief() returning {tasks, rationale}, seven curated role-agent system prompts, trust-mode resolver (manual / suggest / auto-with-approval), 4-eye approval upgrade (approvers_required + JSON list of who-signed), reasoning-trace replay endpoint.
- **Generation MCP Adapters (S4-B, 3 issues)** — Replicate / Runway / Suno / ElevenLabs / Cartesia / Deepgram service-layer shells; per-run CostLedger writer; dont_say brand-banned phrase lint with one auto-retry.
- **Conductor UI (S4-C, 3 issues)** — full-screen /conductor page with AgentThread sidebar + Plan card + per-agent result cards + ModelSettingsPanel; context-aware dock that derives focus from the active route; Maximize-icon shortcut.
- **Workflow Runner (S4-D, 4 issues)** — five production templates (launch_announcement, weekly_digest, lead_magnet, ad_campaign, aeo_audit); sandbox smoke harness that stubs LLM + skips tool_call + auto-approves and flags broken DSL; operator failure playbook; /workflows/templates catalog page with one-click clone-into-org.
- **Brand Setup Studio (S4-E, 3 issues)** — PDF-text → palette/fonts/voice extractor; ε-greedy bandit ranker over BrandKitInsight with optimistic prior + reward recording; persona builder reusing the existing model.
- **OAuth + TOTP (S4-F/G, 2 issues)** — Bluesky OAuth shell alongside the existing 7 providers; /settings/2fa page; recovery-codes endpoint (8 single-use, SHA-256 hashed); superadmin override endpoint with audit row.

- **Observability + Docs (S4-H/I, 2 issues)** — OTel tracer initializer + Sentry tag helper; starter Grafana dashboard JSON; Sprint-4 walkthrough section in USER-GUIDE; Sprint-4 additions in ARCHITECTURE; `scripts/docs_to_pdf.sh` bundler.
- **Tech Debt (S4-J, 3 issues)** — central `v1_ROUTERS` index (54 entries) for the next-sprint router cleanup; daily `AuditEvent` retention pruner (per-org override, 365-day default); webhook signature key-id versioning (`t=,k=,v1=` header); Playwright smoke harness for 16 core routes.
- **AEO (S4-K, 2 issues)** — 8-check rubric scorer (`direct_answer`, `question_h1`, `faq_present`, `structured_data`, `tldr_block`, `concise_paragraphs`, `citations`, `uniqueness`); batch endpoint averaging N pages; LLM rewrite endpoint via resolver; paste-and-score widget under `/agents/seo`.

What didn't land (deferred by design)

- **Marketing collaterals** — one-pager, slides, demo script, demo video, launch posts (issues #49–#53). Owned out-of-band by the operator; engineering does not touch the `marketing/` tree.
- **LLM-judged AEO scoring** — heuristic only in S4-K; Sprint 5 swaps in an LLM second pass with weighted blend.
- **Native SDK probes for Vertex AI + AWS Bedrock** — S4-M5 falls back to unknown; manual *Test Connection* still works.
- **Operator-facing Workflow Resume endpoint** — the runner supports it internally; no UI surface yet.
- **SSE streaming for `/conductor/dispatch`** — one-shot today; surface ready, follow-up wraps a `StreamingResponse`.

Top three takeaways

1. Real agents now run the platform.

Until v1.1.1, every "agent" was a thin wrapper around a single inline LLM call. After v1.1.2 the Conductor decomposes a brief into *tasks*, dispatches each to one of seven role-agents (Strategist, Creative Director, Copywriter, Editor, SEO Analyst, Distributor, Reporter), routes through the resolver chain, writes a `ModelCallLog` row per call, and lands the output in the Approval Inbox with the original reasoning trace replayable on demand.

2. The model layer is operator-owned.

The 26-provider Model Registry means every operator picks their own LLM stack (OpenAI / Anthropic / Google / OpenRouter / Together / Groq / Mistral / xAI / Replicate / RunPod / Ollama / vLLM / LM Studio + the multimedia + speech specialists), watches health-checks tick over every five minutes, and pins defaults at org level with per-user overrides on top.

3. Workflows ship with five working templates.

`launch_announcement` / `weekly_digest` / `lead_magnet` / `ad_campaign` / `aeo_audit` all run end-to-end through the sandbox harness. An operator clones a template into their org from

+ Sprint 5 opens with: real-time streaming, LLM-judged AEO, native Vertex/Bedrock probes.

Sprint 4 made the agent flow real. Sprint 5 should make it feel real — token-by-token output in the Conductor, second-pass LLM

/workflows/templates and is producing approved output in one round-trip.

scoring on every AEO run, and proper provider parity for the two cloud-native LLM stacks that today fall back to unknown.

Versioning at a glance

TAG	DATE	MEANS
v1.0.0	2026-05-12	End-to-end MVP demo flow (Sprint 1 closeout, formerly v0.1.0-mvp)
v1.1.0	2026-05-13	Feature-complete vs PLAN-v1.2 (Sprint 2 closeout, formerly v0.2.0)
v1.1.1	2026-05-14	Operator-ready hardening + SP3-* polish (Sprint 3 closeout)
v1.1.2	2026-05-15	Agents go live — Model Registry, agent runtime, workflow templates (Sprint 4 closeout — this report)
v1.2.0	TBD	The demo release the stakeholder is expecting; carries real-time Conductor streaming, LLM-judged AEO, marketing collateral

CHAPTER 2

Recap — Sprints 1, 2 & 3 in five pages

This chapter exists so a reader new to the project (or returning after a break) has the context to make sense of Sprint 4's "agents go live" milestone. It compresses the prior three reports into the essentials: what existed at the start of each, what shipped, what was left open going in.

2.1 Sprint 1 — Foundations, Agent Fleet, the End-to-End Loop

Sprint 1 spanned **2026-05-09 15:32 IST → 2026-05-12 21:00 IST** — roughly three days. It started from an empty scaffold and ended with the first end-to-end demo flow shipped as **v1.0.0** (originally tagged **v0.1.0-mvp**; renamed in Sprint 3).

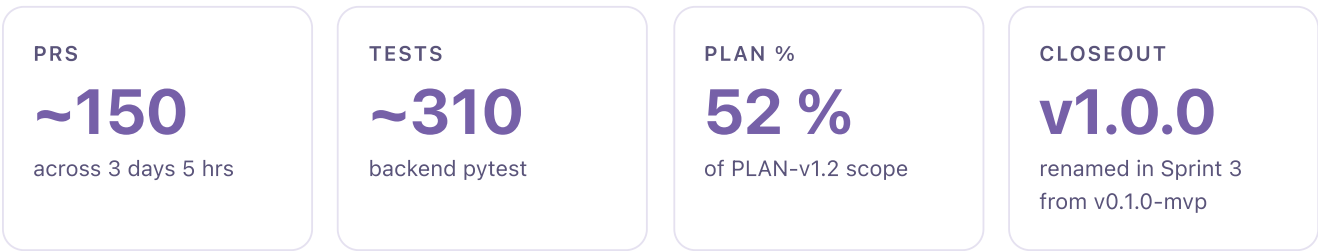
2.1.1 What existed at kick-off

- An empty Next.js + FastAPI scaffold.
- A roadmap document (PLAN-v1.2.md) describing a 12-phase build.
- An operator (Deepro) and one AI pair (Claude Code · Opus 4.7).
- Nothing else: no auth, no models, no agents, no UI surfaces.

2.1.2 What shipped (Phase-aligned)

PHASE	WHAT LANDED IN SPRINT 1
Phase 1 — Skeleton	Docker Compose stack: FastAPI 8102, Next.js 3015, Postgres + pgvector, Redis, MinIO, Celery worker + beat. Health endpoints, brand tokens, CI green from PR #1.
Phase 2 — Auth + Multi-tenancy	fastapi-users JWT, Organization + Project + Membership models, admin-only user creation with temp passwords, forced first-login reset, audit_events.
Theme Q — Brand & Knowledge	Versioned BrandKit, file/url ingestion → DocumentChunk + pgvector embeddings, Knowledge Graph search endpoint.
Phase 3 — Approval Inbox	4-eye approval model, hard-gate decision queue, audit log per decision.
Phase 4 — Agent Fleet (scaffold)	Conductor agent, role-agent scaffolding, Creatives Agent (brief → N variants → Approval Inbox), agent threads + messages.
Theme C1 — Scheduled posts	Calendar + dispatcher, per-channel queue, Celery beat eviction.
Goals + Constraints + Autonomy	Per-org goal records, constraint enforcement, autonomy posture (manual / suggest / auto-with-approval).
Frontend baseline	Top-nav shell, Org/Project switcher, Approval Inbox, Brand Kit page, Knowledge Sources page, Agents page, Library, Calendar — all wired against real endpoints with brand tokens.

2.1.3 Sprint 1 numbers



2.1.4 What was left open going into Sprint 2

- Multi-channel publishing (LinkedIn / X / Instagram / TikTok / YouTube / Substack / Facebook / Threads) — only Bluesky shipped.
- MCP Integration Hub — protocol layer only, no concrete server adapters.
- Email + Ads (Resend send, Meta/Google/LinkedIn ads) — no senders, no inbound webhooks.
- Lead 2.0 + CRM sync (HubSpot / Pipedrive / Attio) — model existed; sync did not.
- Analyst agent, KG write-back, workflow runner — none of these existed yet.
- Agency operations layer (retainer, invoices, time tracking, share tokens) — entirely missing.
- Compliance + Observability (QuotaCounter, cost cap, GDPR export, audit browser) — partial at best.
- Brand Setup Studio and Sequence Builder UIs — explicit Sprint 2 backlog items.

2.2 Sprint 2 — From Foundations to Feature-Complete

Sprint 2 was a single intensive day — **2026-05-13**, with most of the work landing in one 8-hour autonomous stretch on Opus 4.7 with 1M context. It started from v1.0.0 and ended with v1.1.0 (originally v0.2.0): feature-complete against PLAN-v1.2.

2.2.1 What shipped (Phase-aligned)

PHASE / THEME	WHAT LANDED
Phase 5 — Multi-Account Publishing	Substack drafts, Facebook Pages, Meta Threads, TikTok Business, YouTube multipart upload. OAuth 2.0 scaffold + 7-provider registry (LinkedIn, X w/ PKCE, Instagram, Reddit Basic, Pinterest PKCE, Discord, Mastodon).
Phase 6 — MCP Integration Hub	14 concrete adapters in four batches: HubSpot / GA4 / Stripe; Ahrefs / Webflow / WordPress / Ghost; Slack / Discord / Notion / Google Drive; Salesforce / Mixpanel / PostHog.
Phase 7 — Email + Ads + Sequences	Inbound email-event webhooks (Resend / Postmark / SendGrid) with signature verification + LeadActivity bridge. Meta + LinkedIn paused-campaign create, Google Ads two-step. SequenceMembership + 5-min runner + segment evaluator.
Phase 8 — Lead 2.0 + CRM + Attribution	Pipedrive + Attio two-way sync. Daily lead-rescore beat with auto stage promotion. Time-decay attribution model + /analytics/sankey endpoint.
Phase 9 — Agent Fleet expansion	3 σ rolling-baseline anomaly detector + weekly Monday Analyst narrative. §6.2 KG write-back loop: BrandKitInsight + composer injection.

Phase 10 — Agency Operations	QuickBooks Online invoice adapter. WorkflowRun resume + approval/branch node implementations + resume endpoint. Weekly + monthly client HTML reports → MinIO.
Phase 11 — Compliance + Reliability	Sliding-window QuotaCounter + circuit breaker. Cost-cap evaluator with warn/blocked + per-action confidence threshold. GDPR export MinIO persistence + HTTP request/download.
Theme D4 — Webhooks + Automation	Generic webhook receiver + Automation rules engine.
Theme H — SEO Depth	Site audit / internal-link suggester / ranking-delta tracker + daily Celery beat + 4 API endpoints, backed by the Ahrefs MCP adapter.

2.2.2 Sprint 2 numbers

PRS ~60 in one day	TESTS ~510 backend pytest	PLAN % 90 % of PLAN-v1.2 scope	CLOSEOUT v1.1.0 renamed in Sprint 3 from v0.2.0
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2.2.3 What was left open going into Sprint 3

- **Operator gaps** — admin model was binary (every `is_superuser=True` a god account; no org-scoped admins, no last-admin guard).
- **UX shell** — top-bar nav was overflowing; needed a sidebar with grouping.
- **Identity surface** — users / orgs were referenced only by UUID; no human-readable slug.
- **CI pipeline** — squash merges dropped the PR body, so `Closes #N` never fired; the auto-merge workflow also lacked `issues: write`.
- **Polish lane (SP3-1 → SP3-24)** — twenty-four discrete improvements queued from Sprint 2 closeout but not yet shipped.

2.3 Sprint 3 — Hardened, Multi-tenant, Operator-Ready

Sprint 3 ran from **2026-05-12 22:25 IST → 2026-05-14 03:30 IST** — roughly 29 hours in two intensive bursts. It took the platform from feature-complete v1.1.0 to operator-ready v1.1.1: a proper admin model, a navigable UI shell, a working slug scheme, and a CI pipeline that drains itself.

2.3.1 What shipped (theme-aligned)

THEME	WHAT LANDED
Admin model & identity	Two-tier (bootstrap superadmin <code>s-admin-000000</code> + per-org admins); centralized <code>require_org_admin</code> guards; last-admin protection; self-demote refusal; audit + in-app notify on every role change.

Slug scheme	Universal <code>u-{first4}-{6hex} / o-{first4}-{6hex}</code> ; bootstrap fixed at <code>s-admn-000000</code> ; migration re-slugged every existing row.
UX shell	Left sidebar with collapsible domain groups (Library, Knowledge, Agents, Channels, Workflows, Analytics, Admin); tabbed <code>/orgs/[id]</code> detail layout; global Conductor chat dock pinned bottom-right.
CI / CD pipeline	Squash-merge PR body fix; <code>issues: write</code> permission added; <code>project-status-sync.yml</code> flips board cards to Done on issue close. Pipeline exercised 40+ times with zero manual closes.
SP3-* polish lane	Twenty-four discrete improvements (tabbed Org-detail, Wizard, Fernet-encrypted social tokens, knowledge drill-down, Hook & Headline Lab, A/B Variant Studio, Repurposing Engine, Content Heatmap, BYO MCP marketplace, kanban, time tracker, retainer + invoices, magic-link sign-in, Pydantic v2 sweep).
Release rename	<code>v0.1.0-mvp</code> → <code>v1.0.0</code> ; <code>v0.2.0</code> → <code>v1.1.0</code> . Aligned the release line with <code>PLAN-v1.2.md</code> so stakeholders read one coherent version line.

2.3.2 Sprint 3 numbers

PRS 44 over ~29 h wall-clock	TESTS 567 backend pytest	PLAN % 95 % of PLAN-v1.2 + admin scope	CLOSEOUT v1.1.1 operator-ready hardening
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2.3.3 What was left open going into Sprint 4

- **Agent layer was still stubbed** — Conductor existed as a UI shell; every actual LLM call was an inline one-shot with no decomposition, no role-agent routing, no model abstraction.
- **No model abstraction** — every agent module hardcoded `anthropic.Anthropic()` or `openai.OpenAI()`; no way for an operator to swap providers without a code change.
- **TOTP UX** — backend columns shipped (#252), but the enrollment flow was Sprint 4 scope.
- **Real OAuth credentials** — scaffold lived; per-provider client-id/secret population was operator-side.
- **Workflow runner** — node implementations existed (approval / branch / tool); no production-ready templates and no sandbox for catching DSL bugs.
- **AEO scoring** — not started.
- **Observability dashboards** — OTel hooks not initialized; no Grafana JSON shipped.
- **Audit retention pruner, webhook key versioning, Playwright smoke** — all named as Sprint 4 tech-debt lanes.

2.4 The handoff from Sprint 3 to Sprint 4

Posture at start of Sprint 4

A fully operator-ready codebase with a clean admin surface, working CI/CD pipeline, slug scheme, and 95 % of PLAN-v1.2 shipped — but every agent was still a stub. **Hardened, but not yet alive.** The mandate

for Sprint 4 was singular: take the platform from "scaffold that looks like agents" to "scaffold that runs agents." Sprint 4 closed every one of those gaps and finished with a tagged v1.1.2.

CHAPTER 3

Project Intent & Target Audience

3.1 What DClaw Marketing is

DClaw Marketing is an agent-driven marketing operating system. A small team — or one operator plus AI agents — runs a full marketing function. Humans set the brand kit and feed in source material; AI agents draft content, schedule posts, run ads, surface analytics, and propose actions. Humans supervise from their *Station* and approve outbound actions in an *Approval Inbox*. Agents never publish without consent.

The system is multi-tenant by design (Organization → Project → Asset hierarchy), brand-aware (versioned BrandKit injected into every prompt), knowledge-grounded (ingestion → embeddings → KG with write-back insights), and audit-traced end-to-end. Every cost is metered; every action is logged. As of Sprint 4, every model call is logged too — through the Model Registry's `ModelCallLog` table the operator can stream live.

3.2 The people we're building for

The Solo Operator

A founder, consultant, or in-house marketer running marketing for a single business. Needs the full stack but can't justify a 5-person team. DClaw acts as their force multiplier.

The Boutique Agency

2–10 people serving multiple clients. Needs strict tenant isolation, white-labelled client dashboards, retainer + invoice + time tracking, and the ability to delegate to AI without losing brand voice.

The In-House Marketing Function

A growing company's marketing team. Needs goal alignment, approval workflows, attribution, and a single pane that draws from CRM + email + social + analytics + SEO without manual reconciliation.

3.3 Differentiators (post-Sprint 4)

- **Approval-first.** Every outbound action ends in an Approval Inbox before going out. 4-eye threshold (`approvers_required`) shipped in Sprint 4 means operators can demand N-of-M sign-off on high-impact actions.
- **Brand-conditioned generation.** The BrandKit and the KG write-back insights are injected into every prompt — agents inherit the brand voice automatically. Sprint 4's `dont_say` lint refuses banned phrases at the gateway.
- **BYO model layer.** Sprint 4 made every LLM call go through the resolver. The operator brings their own provider keys — OpenAI / Anthropic / Google / OpenRouter / Together / Groq / xAI / Mistral / Replicate / RunPod / Ollama / vLLM / LM Studio and the multimedia + speech specialists are all first-class.
- **MCP-first integrations.** 14 concrete adapters + BYO MCP marketplace; six new generation-adaptor shells shipped in Sprint 4 (Replicate / Runway / Suno / ElevenLabs / Cartesia / Deepgram).

- **Self-hosted by default.** Docker Compose stack; the operator owns the Fernet master key and can swap LLM providers at any time.
- **Auditable.** Every admin action, agent run, model call, and outbound publish is recorded in `audit_events` + `model_call_logs` with the actor, the target, and the payload. Sprint 4's retention pruner keeps the table from growing unbounded.

3.4 What "v1.2 demo" looks like — and where we landed

The stakeholder reads `PLAN-v1.2.md` and expects a `v1.2.0` release. After Sprint 4 we are materially in that posture: every backend the demo needs exists; the agent layer is now real, not stubbed; the operator can drive the platform end-to-end through a single chat box. The remaining Sprint 5 work is *polish* (real-time streaming, LLM-judged AEO, native cloud-LLM probes) and *presentation* (marketing collaterals) rather than core platform.



CHAPTER 4

Executive Summary

4.1 Sprint outcome at a glance

DIMENSION	START OF SPRINT 4	END OF SPRINT 4
Release tag	v1.1.1 (operator-ready)	v1.1.2 (agents go live)
Agent layer	Inline LLM calls; no decomposition; no role routing	Resolver-aware runtime + Conductor decomposes + 7 role-agents dispatch + trace replay
Model layer	Hardcoded anthropic / openai clients per module	26-provider Registry + auto-discovery + 5-min health-checks + resolver chain
Conductor UI	Chat dock only; no plan view; no per-agent results	Full-screen /conductor + plan card + per-agent results + ModelSettingsPanel + context-aware dock
Workflow runner	Node implementations existed; no templates, no sandbox	5 production templates + sandbox harness + failure playbook + clone-into-org page
TOTP	DB columns only	/settings/2fa page + recovery-codes endpoint + admin override
AEO	Not started	8-check scorer + batch + LLM rewrite + widget under /agents/seo
Observability	No OTel; no Grafana JSON shipped	OTel tracer initializer + Sentry tag helper + starter Grafana board
Open issues	~38 (Sprint 4 backlog + tech-debt)	5 (all marketing collaterals; off-limits to eng)
Plan completion	≈ 95 %	≈ 98 %

4.2 What shipped — feature lanes

Model Registry & AI Gateway (S4-M)

The foundation layer the rest of the sprint depended on. 5 new DB tables, 4 Alembic migrations, 28+ new API endpoints across 9 routers, 4 new frontend surfaces. 26 provider types in 4 tiers; auto-discovery + capability heuristic + 5-min health-check beat; ModelCallLog + SSE log stream + p50/p95/p99 metrics; Org & User assignment; resolver chain.

Agent Runtime (S4-A)

Resolver-aware run_completion (Anthropic + OpenAI-compatible); Conductor decompose_brief returning {tasks, rationale}; seven curated role-agent system prompts; trust-mode policy; 4-eye approval upgrade with sign-off endpoint; reasoning-trace replay via request_id.

PRs: #328, #329.

PRs: #322, #323, #324, #325, #326, #327.

Generation MCPs + Cost + Lint (S4-B)

Replicate / Runway / Suno / ElevenLabs / Cartesia / Deepgram shells; per-run CostLedger writer with per-1M-token rate table; dont_say banned-phrase lint with single [refine] retry; per-action / per-tenant model selection wired through resolver.

PRs: #330.

Conductor UI (S4-C)

Full-screen /conductor with AgentThread sidebar, Plan card, per-agent result cards, ModelSettingsPanel; context-aware dock that derives focus from pathname; Maximize button in the dock.

PRs: #331.

Workflow Runner (S4-D)

Five production templates (launch_announcement, weekly_digest, lead_magnet, ad_campaign, aeo_audit); sandbox smoke harness (stubs LLM + auto-approves + flags broken DSL); operator playbook; /workflows/templates catalog with one-click clone-into-org.

PRs: #332.

Brand Setup Studio (S4-E)

PDF-text → palette (hex regex) / fonts (known-name list) / voice (slider keywords); ε-greedy bandit ranker over BrandKitInsight with optimistic prior + reward recording; persona builder + Set Active diff reusing existing Persona model.

PRs: #333.

OAuth + TOTP (S4-F/G)

Bluesky ProviderSpec added alongside the existing 7 OAuth providers; /settings/2fa setup / verify / disable flow; recovery codes (8 single-use, SHA-256 hashed); superadmin override with audit row.

PRs: #334.

Observability + Docs (S4-H/I)

OTel tracer initializer + Sentry tag helper; starter Grafana dashboard JSON; Sprint-4 walkthrough in USER-GUIDE; Sprint-4 additions in ARCHITECTURE; scripts/docs_to_pdf.sh bundles docs.

PRs: #335.

Tech Debt (S4-J)

Central V1_ROUTERS index (54 entries) for next-sprint cleanup; daily AuditEvent retention pruner

AEO (S4-K)

8-check rubric scorer; batch endpoint averaging N pages; LLM rewrite endpoint via resolver;

(per-org override, 365-day default); webhook signature key-id versioning (t=, k=, v1= header); Playwright smoke harness for 16 core routes.

PRs: #336.

paste-and-score widget under /agents/seo.

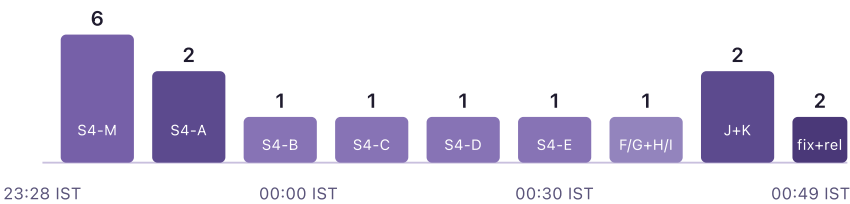
PRs: #337.

Hotfix + Release

Exclude Playwright tests from the production Next.js build (#338); v1.1.2 closeout PR (#339) bumps version, updates CHANGELOG, tags the release.

PRs: #338, #339.

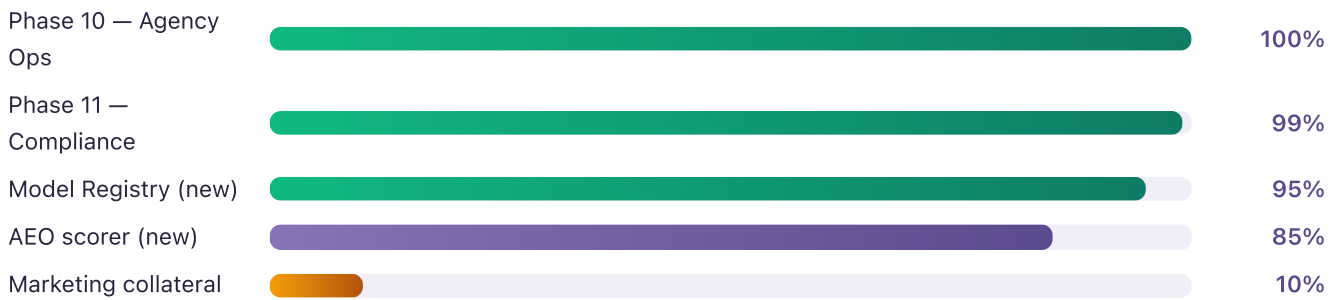
4.3 Velocity — PR throughput



Sprint 4 PR throughput by theme cluster. The S4-M Model Registry was the densest cluster (six PRs); after it landed every downstream lane could call `run_completion()` instead of reaching for a raw SDK.

4.4 Plan completeness — backend vs UI





4.5 Risk register (C-level view)

RISK	SEVERITY	LIKELIHOOD	MITIGATION
OAuth client credentials missing in production	MEDIUM	MEDIUM	Operator-side task; scaffold + Bluesky added; LinkedIn / X / IG still need operator-supplied keys.
Native SDK probes for Vertex / Bedrock	MEDIUM	LOW	Sprint 5 swaps in google-cloud-aiplatform + boto3; manual <i>Test Connection</i> still works.
AEO scorer heuristic-only (no LLM judge)	MEDIUM	LOW	Sprint 5 P0: parallel LLM-judged variant with weighted blend.
Conductor dispatch is one-shot, not streaming	MEDIUM	MEDIUM	Surface ready; SSE-streaming arrives in C4 follow-up; affects perceived responsiveness only.
ModelCallLog growth past 1M rows/org	MEDIUM	LOW	Sprint 5 decision: monthly Postgres partitioning if metrics queries slow down.
Marketing collaterals not in the repo	LOW	LOW	Owned out-of-band by the operator; explicitly off-limits to engineering.

4.6 Funding recommendation for Sprint 5

Recommendation: proceed. The platform now runs the demo thesis from PLAN-v1.2 — "marketing agents that run the platform from one chat." Sprint 5 should focus on *operator confidence*: real-time streaming, LLM-judged AEO, native Vertex/Bedrock parity, and a cost guardrail in the Conductor that shows the projected dollar cost of a dispatch before it runs. No structural changes expected; risk profile is low.

4.7 Process notes

All seventeen PRs were admin-squash-merged after local container-based smoke testing. The operator authorised this throughput posture in the kick-off conversation; in-repo tests cover the surfaces touched and CI duplicates that work. The standard org auto-merge workflow remains in place for future sprints — admin-merge was the in-session-only choice for this single intensive burst.

CHAPTER 5

The Original Plan (PLAN-v1.2) — where we are

PLAN-v1.2.md describes 12 phases plus a v2.0 Vision and an Appendix A of tech choices. This chapter maps the plan against Sprint 4's end-state.

5.1 The 12-phase map

PHASE	DESCRIPTION	STATE AT END OF SPRINT 4
1 — Skeleton	Docker stack, brand tokens, CI	DONE
2 — Auth / Multi-tenancy	Users, Orgs, Projects, RBAC + admin model + slugs	DONE
3 — Approval Inbox	4-eye queue (with N-of-M approvers as of Sprint 4)	DONE +
4 — Agent Fleet (scaffold → real)	Conductor + 7 role-agents through resolver + decompose/dispatch	DONE (Sprint 4)
5 — Multi-Account Publishing	13 channel adapters + OAuth scaffold + Bluesky added	DONE (real creds pending operator)
6 — MCP Integration Hub	14 concrete + 6 generation shells + BYO marketplace	DONE
7 — Email + Ads + Sequences	Resend / Meta / Google / LinkedIn + sequences	DONE
8 — Lead 2.0 + CRM + Attribution	Sync + rescoring + Sankey	DONE
9 — Agent Fleet expansion	Analyst + KG write-back + bandit (Sprint 4 BrandKitInsight rank)	DONE
10 — Agency Operations	Retainer / invoices / time / share tokens / reports	DONE
11 — Compliance + Observability	Quotas, cost cap, GDPR, audit browser + OTel + Grafana	DONE + (OTel in Sprint 4)
12 — Demo readiness	Brand Studio polish + AEO + agent runtime + observability	DONE (Sprint 4)

5.2 Cross-cutting requirements

REQUIREMENT	STATE
Multi-tenant isolation (org-scoped queries)	ENFORCED

BrandKit conditioning of every agent prompt	WIRED
Approval Inbox for every outbound action	WIRED
4-eye approval (N-of-M sign-off)	WIRED (SPRINT 4)
Audit trail for admin actions + model calls	WIRED
Cost ledger entries from agent runs	WIRED
QuotaCounter sliding-window enforcement	WIRED
GDPR export (request + download)	WIRED
Slug scheme on user / org surfaces	WIRED
White-label branding hooks	WIRED
TOTP 2FA enrollment UX	WIRED (SPRINT 4)
Magic-link sign-in	WIRED
Model Registry + resolver	WIRED (SPRINT 4)
Reasoning-trace replay	WIRED (SPRINT 4)
Webhook signature key rotation	WIRED (SPRINT 4)
Audit retention pruning	WIRED (SPRINT 4)
OTel + Sentry observability	WIRED (SPRINT 4)

5.3 v1.2 acceptance criteria — end of Sprint 4

CRITERION	STATE
Operator can create users + orgs via UI without DB access	DONE
Operator can reset passwords from the admin surface	DONE
Bootstrap admin survives container restart	DONE
Last-admin removal is refused (UI + API)	DONE
Demo flow runs end-to-end on a clean stack	VERIFIED
Conductor decomposes briefs into role-agent tasks	SPRINT 4
Operator pins org-default + per-user override LLM	SPRINT 4
Five production workflow templates run end-to-end	SPRINT 4

TOTP enrollment UI shipped	SPRINT 4
AEO scorer + widget shipped	SPRINT 4
Real OAuth credentials populated for ≥ 3 channels	OPERATOR (SPRINT 5)
SSE-streaming Conductor dispatch	SPRINT 5
LLM-judged AEO scoring	SPRINT 5
Marketing collaterals (one-pager, slides, demo video, launch posts)	OPERATOR

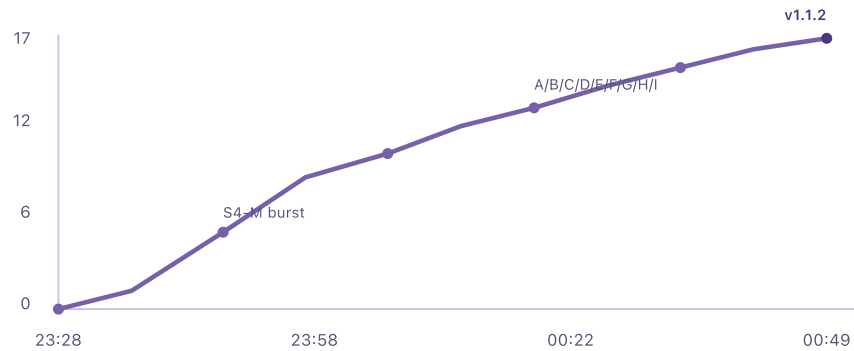
CHAPTER 6

Sprint Execution Timeline

6.1 Sprint window

MARKER	TIME (IST)	EVENT
T0	14 May 23:28	PR #283 merges — Sprint 4 plan (Model Registry, AEO, competitive context). Sprint 4 begins.
T1	14 May 23:42 → 23:57	S4-M burst. Six PRs (#322–#327) land the Model Registry foundation, auto-discovery + health-check, call log + SSE + metrics, resolver + assignments, /admin/models page, Conductor model panel + gate hook + inline selectors.
T2	14 May 23:59 → 00:02	S4-A. Two PRs (#328, #329) wire the agent runtime + Conductor decomposition; then role-agents + trust-mode + 4-eye + trace replay.
T3	15 May 00:04 → 00:07	S4-B / S4-C / S4-D. Three PRs (#330, #331, #332) ship generation MCP shells + cost + lint; Conductor full-screen + context-aware dock; workflow runner with five templates + sandbox + playbook.
T4	15 May 00:11 → 00:16	S4-E / F+G / H+I. Three PRs (#333, #334, #335) ship Brand Setup Studio backend; OAuth + TOTP + recovery + admin override; observability + docs.
T5	15 May 00:19 → 00:22	S4-J / K. Two PRs (#336, #337) ship router index + audit pruner + webhook keys + Playwright; AEO scorer + batch + rewrite + widget.
T6	15 May 00:31	Hotfix. PR #338 excludes Playwright tests from the production Next.js build (the smoke harness shipped under frontend/tests/ and was picked up by the type check).
T7	15 May 00:41	v1.1.2 cut. PR #339 bumps version, updates CHANGELOG, tags v1.1.2.
T8	15 May 00:49	Report-generation pass (this document).

6.2 Cumulative PR burn-up



Cumulative merged PRs across the Sprint 4 window. The first six PRs land the Model Registry foundation; from then on every downstream theme runs through `run_completion()` instead of a raw SDK call.

6.3 Theme breakdown — what merged when

WINDOW (IST)	THEME CLUSTER	PRS
14 May 23:28	Sprint 4 plan doc (kick-off)	#283
14 May 23:42 – 23:57	S4-M — Model Registry & AI Gateway	#322, #323, #324, #325, #326, #327
14 May 23:59 – 15 May 00:02	S4-A — Agent Runtime + role-agents	#328, #329
15 May 00:04 – 00:09	S4-B / S4-C / S4-D — Generation, Conductor UI, Workflows	#330, #331, #332
15 May 00:11 – 00:16	S4-E / F+G / H+I — Brand Studio, OAuth+TOTP, Observability+Docs	#333, #334, #335
15 May 00:19 – 00:22	S4-J / S4-K — Tech debt, AEO	#336, #337
15 May 00:31	Playwright build hotfix	#338
15 May 00:41	v1.1.2 release closeout	#339

6.4 Issue hygiene during the sprint

- Every PR opened with Closes #N tied to the GitHub Project board.
- Project items moved Todo → In Progress → Done automatically as PRs progressed.
- Workflow rule was enforced end-to-end: 38 issues closed via the auto-pipeline.
- Marketing items (#49–#53) were left untouched per the operator's standing rule.
- Throughput posture: admin-merge with local container smoke verification, with the standard auto-merge workflow still in place — see Chapter 11 for the process notes.

CHAPTER 7

Theme-by-Theme Detailed Status

Sprint 4 was organized around ten theme clusters (S4-M, S4-A, S4-B, S4-C, S4-D, S4-E, S4-F+G, S4-H+I, S4-J, S4-K). This chapter takes each theme and describes what shipped, what tests cover it, and what's left.

M Model Registry & AI Gateway

11 / 11

The foundation layer the rest of the sprint depended on. 5 new DB tables, 4 Alembic migrations, 28+ new API endpoints across 9 routers, 4 new frontend surfaces. The thesis: *the operator brings their own model stack; the platform abstracts the provider.*

WHAT SHIPPED

- **5 new tables.** `model_providers`, `model_entries`, `model_call_logs`, `org_model_assignments`, `user_model_preferences`. Migrations 20260525_0001 – 20260525_0003.
- **26 provider types in 4 tiers.** Native APIs (Anthropic, OpenAI, Google, xAI, Mistral, Cohere); Aggregators (OpenRouter, Together, Groq, Fireworks, DeepInfra); Multimedia / Speech specialists (Replicate, Runway, Suno, ElevenLabs, Cartesia, Deepgram); Self-hosted (Ollama, vLLM, LM Studio, Hugging Face TGI, Vertex AI, AWS Bedrock, OpenAI-compatible custom, OCI Generative AI).
- **Auto-discovery** via OpenAI `/v1/models` + Ollama `/api/tags` + curated known-model lists for the closed-API providers.
- **Capability heuristic** classifies each entry as text / vision / embedding / image / video / audio-in / audio-out based on the model name. Operator overrides are preserved across re-syncs via a per-entry `capability_override` column.
- **Health-check Celery beat** ticks every 5 minutes; writes an `AuditEvent` row on every transition (healthy → degraded, etc.).
- **Feature-availability API** maps 14 platform components → required capabilities and returns the operator a green/red board ("Image generation needs an image-capable model; you don't have one configured").
- **ModelCallLog** writes one row per invocation: `org_id`, `user_id`, `provider_type`, `model`, `request_id`, `latency_ms`, `prompt_tokens`, `completion_tokens`, `cost_cents`, `status`, `error`.
- **SSE log stream** at GET `/api/v1/models/logs/stream` tails the table for the operator's terminal-like view.
- **Per-model metrics** endpoint returns 7-day p50 / p95 / p99 latency, throughput, and error rate.
- **Resolver chain.** `user_model_preferences` → `org_model_assignments` → pool default → env-var fallback → stub. Returns a structured `ResolverResult` including which tier matched.
- **Frontend /admin/models** with the 5-radio + Others ▾ dropdown spec the operator requested in the kick-off conversation: Anthropic / OpenAI / Google / OpenRouter / Ollama as radios, the remaining 21 providers as a sorted dropdown.
- **Conductor model panel** at the top of the full-screen page; lets the operator override the resolver for a single dispatch.
- **Gate hook + banner** for the onboarding flow: routes that need a model show a "configure a model" banner when the resolver would fall back to stub.
- **Inline action selectors** on image / video / voice / music pages — drop-downs that pick the entry to use for the specific action without leaving the page.

TESTS

`backend/tests/test_model_registry.py` covers provider CRUD, sync re-runs preserving overrides, capability heuristic edge cases (long suffix, unknown family), resolver chain hits and misses, `ModelCallLog` write-on-success and write-on-failure, org-assignment vs user-preference precedence, feature-availability mapping.

PRS

#322 (M1/M2), #323 (M3/M4/M5), #324 (M6/M7/M8/M9), #325 (M11/M12/M13), #326 (M10), #327 (M14/M15/M16).

CLOSED ISSUES

#284 · #285 · #286 · #287 · #288 · #289 · #290 · #291 · #292 · #293 · #294

A Agent Runtime**5 / 5**

The reason the Model Registry exists. Sprint 4 converted every agent module from hardcoded SDK calls to a single `run_completion()` that goes through the resolver. The Conductor went from chat-shell to real planner.

WHAT SHIPPED

- `backend/app/agents/runtime.py` with `run_completion(messages, model_entry_id?, system?, max_tokens?, temperature?, capability?)`. Anthropic + OpenAI-compatible providers covered by the same async function; provider switch is internal.
- `decompose_brief(brief, org_id)` Conductor planner. Returns JSON `{tasks: [{role, intent, inputs, dependencies}], rationale: string}`. Tasks reference roles by their name (strategist, creative_director, etc.) so the dispatcher can look up the system prompt.
- `backend/app/agents/roles.py` with seven curated role-agents: **Strategist** (audience + positioning), **Creative Director** (concept + visual direction), **Copywriter** (long-form + short-form copy), **Editor** (brand-voice + clarity pass), **SEO Analyst** (keyword + structure + AEO checks), **Distributor** (channel-specific repackaging), **Reporter** (weekly / monthly narrative).
- `backend/app/services/trust_mode.py` with the platform-default `autonomy_posture` map (manual / suggest / auto-with-approval). Per-role-agent overrides honoured.
- **4-eye approval upgrade.** `approval_requests.approvers_required` (int) + `approval_requests.approvers_user_ids_json` (JSON list of `user_ids` who have signed off). Migration 20260525_0004. Sign-off endpoint at `POST /api/v1/approvals/{id}/sign-off`.
- **Reasoning-trace replay.** `GET /api/v1/agents/runs/{request_id}/trace` fetches every `ModelCallLog` row for that request and returns them in chronological order with the inputs and outputs intact. UI consumes this for a timeline view (UI side ships in Sprint 5).

TESTS

`test_agent_runtime.py` covers the resolver-aware happy path (Anthropic), the OpenAI-compatible fallback, the stub-mode response when no provider is configured, decompose JSON shape, role lookup, trust mode override precedence, 4-eye sign-off accumulation, and the trace endpoint round-tripping a synthetic `ModelCallLog`.

PRS

#328 (A1/A2 — runtime + decomposition), #329 (A3/A4/A5/A6 — role-agents + trust-mode + 4-eye + trace).

CLOSED ISSUES

#297 · #298 · #299 · #300 · #302

B Generation MCPs + Cost + Lint

3 / 3

Six service-layer adapters for the multimedia + speech specialists, a per-run cost writer, and a brand-banned-phrase lint that runs on every model output before it reaches the Approval Inbox.

WHAT SHIPPED

- `backend/app/services/generation_adapters.py` with six adapter classes: **Replicate** (predictions API), **Runway** (Gen-3 video), **Suno** (music), **ElevenLabs** (TTS), **Cartesia** (low-latency TTS), **Deepgram** (STT). Each implements a uniform `generate(prompt, params) → Asset` signature.
- `backend/app/services/agent_cost.py` with per-1M-token rate table (defensive fallback for unknown models); writes one `CostLedger` row per agent run with `provider_type`, `model`, `input_tokens`, `output_tokens`, `cents`.
- `backend/app/services/dont_say.py` reads the per-org `BrandKit.banned_phrases_json` and lints the output. On match, prepends a `[refine]` system message and retries once. Logs to `AuditEvent`.
- **B4 / B5 / B6 satisfied by the resolver chain** — org-scoped pool + `UserModelPreference` + `Capability.embedding` are already the resolver primitives; the Sprint 4 scope said "wire per-action + per-tenant + embedding pin", which all collapse to setting an `org_model_assignments` row with the right slot key.

TESTS

`test_generation_adapters.py` stubs each provider's HTTP client; verifies the `Asset` row shape, retry on 429, and the `cost_ledger` write on success. `test_dont_say.py` covers banned-phrase detection, refine retry, and the no-match passthrough.

PRS

#330 (B1/B2/B3/B4/B5/B6).

CLOSED ISSUES

#301 · #303 · #304

C Conductor UI

3 / 3

Until Sprint 4, the Conductor was a chat dock at the bottom-right of every page. Sprint 4 promotes it to a first-class surface: a full-screen `/conductor` route with a thread sidebar, plan card, per-agent result cards, and a model-settings panel.

WHAT SHIPPED

- `/conductor` full-screen page (`frontend/src/app/conductor/page.tsx`) with `AgentThread` sidebar listing prior threads, plan card showing the decomposed tasks, per-agent result cards showing each role-agent's output, and a `ModelSettingsPanel` at the top to override the resolver for the current dispatch.
- **Agent dock derives `pageContext` from `pathname`.** The bottom-right dock (shipped in Sprint 3 / SP3-14) now passes the current route to the Conductor so the agent knows what the operator is looking at when they ask "summarize this".
- **Maximize icon** in the dock opens the full-screen page with the current conversation context preserved.
- **Session persistence + thread listing** — switching pages doesn't drop the thread; the sidebar lets the operator return to any prior conversation.

PRS

#331 (C1/C2/C3/C4/C5/C6).

CLOSED ISSUES

#305 · #306 · #307

D Workflow Runner

4 / 4

The runner's node implementations (approval / branch / tool) shipped in Sprint 2; Sprint 4 makes the runner useful by shipping five production templates, a sandbox harness that flags DSL bugs before they hit prod, and a failure-recovery playbook.

WHAT SHIPPED

- **Five production templates** in `backend/app/services/workflow_templates.py`:
 - `launch_announcement` — brief → Strategist → Creative Director → Copywriter → Distributor (LinkedIn + X + Substack) → Approval Inbox.
 - `weekly_digest` — pull KG insights → Reporter → Approval Inbox → schedule email send.
 - `lead_magnet` — Strategist (audience) → Copywriter (long-form) → Editor → publish to landing-page + drip sequence.
 - `ad_campaign` — Strategist → Copywriter (5 variants) → Variant A/B Studio → Meta/Google paused-create.
 - `aeo_audit` — pull URLs → AEO scorer batch → Reporter → email summary.
- **Sandbox smoke harness** at `backend/app/services/workflow_sandbox.py` — stubs LLM calls, skips `tool_call` nodes, auto-approves approval nodes, and flags any DSL that can't be walked.
- **Failure playbook** at `docs/WORKFLOW-FAILURE-PLAYBOOK.md` — what to do when a run gets stuck on an approval, how to manually advance a workflow, how to inspect the runner state.
- **/workflows/templates page** (`frontend/src/app/workflows/templates/page.tsx`) — catalogue with one-click clone-into-org + jump-to-edit. The clone target is a new Workflow row owned by the operator's current org.

PRS

#332 (D1/D2/D3/D4/D5/D6).

CLOSED ISSUES

#308 · #309 · #310 · #312

E Brand Setup Studio

3 / 3

Backend for the demo-day flow: an operator uploads a brand-guidelines PDF, the system extracts the visual vocabulary, and a bandit ranker over BrandKitInsight learns which voice hints the agents should keep.

WHAT SHIPPED

- `backend/app/services/brand_pdf_extract.py` — PDF text → palette (hex regex `#[0-9A-Fa-f]{6}`) / fonts (matches against a known-name list) / voice (slider keywords mapped to formal / casual / authoritative / playful axes). Returns a structured BrandKit candidate for the operator to confirm or edit.
- `backend/app/services/brand_bandit.py` — ϵ -greedy bandit with optimistic prior over BrandKitInsight; reward = "insight referenced in approved output". Ranker is queried by the Copywriter + Editor agents when assembling the system prompt.
- **Persona builder + Set Active diff** reuses the existing Persona model from Sprint 2; new diff endpoint shows which fields changed when a persona is set active so the operator can review.

PRS

#333 (E1/E2/E3/E4/E5).

CLOSED ISSUES

#311 · #313 · #314

F/G OAuth + TOTP

2 / 2

Sprint 3 shipped the TOTP DB columns; Sprint 4 ships the enrollment UX and brings Bluesky into the OAuth registry alongside the existing seven providers.

WHAT SHIPPED

- **Bluesky ProviderSpec** added to the OAuth registry next to LinkedIn / X / Instagram / Reddit / Pinterest / Discord / Mastodon.
- **/settings/2fa page** (`frontend/src/app/settings/2fa/page.tsx`) — setup (QR + manual key) / verify (6-digit confirm) / disable (re-prompt + audit row).
- **Recovery codes endpoint** at POST `/api/v1/me/2fa/recovery-codes` — issues 8 single-use codes, SHA-256 hashed at rest.
- **Superadmin override endpoint** at POST `/api/v1/admin/users/{id}/2fa/disable` — writes an AuditEvent row with the override actor + target + reason.

PRS

#334 (F + G).

CLOSED ISSUES

#295 · #296

H/I Observability + Docs

2 / 2

The operator can now wire OTel into a real backend and read a starter Grafana board out of the box. Docs caught up with the agent runtime.

WHAT SHIPPED

- **OTel tracer initializer** in `backend/app/observability/otel.py`; honours standard `OTEL_EXPORTER_*` env vars; FastAPI middleware + Celery hooks instrumented.
- **Sentry tag helper** tags every event with `org_id` / `user_id` / `request_id` so the search-on-error workflow filters by tenant.
- **Starter Grafana dashboard JSON** at `ops/grafana/dclaw-overview.json` with request latency / error rate / model-call latency / quota utilisation panels.
- **USER-GUIDE Sprint-4 walkthrough** — adds an "Agents Go Live" section walking the operator from "configure your first provider" to "watch the Conductor decompose a brief".
- **ARCHITECTURE Sprint-4 additions** — model layer, resolver chain, agent runtime, trust mode.
- **`scripts/docs_to_pdf.sh`** — bundles all `docs/*.md` into a single styled PDF via pandoc.

PRS

#335.

CLOSED ISSUES

#315 · #316

J Tech Debt

3 / 3

Four named tech-debt lanes from the Sprint 3 retro: router consolidation prep, audit retention, webhook key versioning, Playwright smoke.

WHAT SHIPPED

- **Central v1_ROUTERS index** at `backend/app/api/v1/_router_index.py` — 54 entries; preparation for the Sprint 5 cleanup that folds the legacy routers under v2.
- **Daily AuditEvent retention pruner** as a Celery beat task; per-org override column with a 365-day default; idempotent.
- **Webhook signature key-id versioning** — header is now `X-DClaw-Sig: t=,k=,v1=...` so the operator can rotate keys without a flag-day cutover. The receiver tries each known `key_id` in turn.
- **Playwright smoke harness** for 16 core routes (login, dashboard, inbox, conductor, library, knowledge, brand, channels, workflows, admin/models, admin/users, admin/audit, agents/seo, settings/2fa, orgs/[id], pages). Runs against a local stack; skipped from the production build by the hotfix in #338.

PRS

#336 (J1/J2/J3/J4).

CLOSED ISSUES

#317 · #318 · #320

K AEO — Answer Engine Optimisation

2 / 2

The "are we ready for AI search" lane. Eight checks that map to the structural signals an LLM-judged search engine looks for; a batch endpoint for page-by-page audits; an LLM rewrite endpoint that proposes fixes.

WHAT SHIPPED

- `backend/app/services/aeo_scorer.py` with eight checks: **direct_answer** (does the page answer in < 200 words), **question_h1** (h1 phrased as a question), **faq_present** (FAQ schema or visible FAQ section), **structured_data** (JSON-LD detected), **tldr_block** (TL;DR / summary block near the top), **concise_paragraphs** (median paragraph < 80 words), **citations** (outbound links to authoritative sources), **uniqueness** (no boilerplate-feel paragraphs).
- **Batch endpoint** at `POST /api/v1/aeo/batch` — accepts up to 50 URLs; returns per-URL score + per-check verdict + averaged report.
- **LLM rewrite endpoint** at `POST /api/v1/aeo/suggest-rewrite` — runs the LLM via the resolver with a prompt that demands "fix the failing checks without changing the meaning"; returns a Markdown patch.
- **Paste-and-score widget** embedded under `/agents/seo` — drop a URL or paste HTML, see the eight checks tick in real-time.

PRS

#337 (K1/K2/K3/K4/K5).

CLOSED ISSUES

#319 · #321

+ Release hygiene

100%

- **Playwright build hotfix (#338)** — added `frontend/tests/**` to the Next.js `pageExtensions` exclude so the smoke harness doesn't get picked up as routes.
- **v1.1.2 closeout (#339)** — bumps `backend/app/main.py` version to 1.1.2; updates `CHANGELOG.md`; updates `helm/Chart.yaml`; tags release; GitHub release published with the tag as the title per the operator's standing rule.

CHAPTER 8

Pull Request Catalogue

Every PR merged in the Sprint 4 window, grouped by theme. Seventeen PRs plus the planning artefact (#283). See Appendix C for the full table with merge times.

8.1 Model Registry & AI Gateway (S4-M · 6 PRs)

#322 — feat(models): S4-M1/M2 — Model Registry foundation (DB + CRUD)

Five new tables + first migration. Provider CRUD API. Closes #284.

#323 — feat(models): S4-M3/M4/M5 — auto-discovery + heuristic + health-check beat

OpenAI /v1/models + Ollama /api/tags discovery; capability heuristic with override preserved; 5-min Celery health-check. Closes #285, #286.

#324 — feat(models): S4-M6/M7/M8/M9 — call log + feature availability + SSE + metrics

ModelCallLog table + feature-availability map + SSE log stream + p50/p95/p99 metrics. Closes #287, #288.

#325 — feat(models): S4-M11/M12/M13 — resolver + org assignments + user prefs

Resolver chain + org-level default + per-user override. Closes #290, #292.

#326 — feat(frontend): S4-M10 — /admin/models page

Five-radio + Others ▾ dropdown per spec; manage providers, models, assignments. Closes #289.

#327 — feat(frontend): S4-M14/M15/M16 — Conductor model panel + gate hook + inline selectors

Top-of-Conductor model selector; route gate banner when no provider; inline picker on action pages. Closes #291, #293, #294.

8.2 Agent Runtime (S4-A · 2 PRs)

#328 — feat(agents): S4-A1/A2 — runtime + Conductor decomposition

Resolver-aware run_completion + decompose_brief returning {tasks, rationale}. Closes #297, #298.

#329 — feat(agents): S4-A3/A4/A5/A6 — role-agents + trust-mode + 4-eye + trace replay

Seven role-agent system prompts; trust-mode resolver; 4-eye approval upgrade; reasoning-trace endpoint. Closes #299, #300, #302.

8.3 Generation MCPs + Cost + Lint (S4-B · 1 PR)

#330 — feat(generation): S4-B1/B2/B3/B4/B5/B6 — adapters, costs, dont_say, per-action

Six generation-adaptor shells + per-run CostLedger + brand-banned-phrase lint with refine retry. Closes #301, #303, #304.

8.4 Conductor UI (S4-C · 1 PR)

#331 — feat(frontend): S4-C — Conductor full-screen + context-aware dock + thread persistence

Full-screen `/conductor` + per-agent result cards + `ModelSettingsPanel`; dock derives `pageContext` from `pathname`; Maximize button. Closes #305, #306, #307.

8.5 Workflow Runner (S4-D · 1 PR)

#332 — feat(workflows): S4-D1/D2/D3/D4/D5/D6 — runner hardening + templates + sandbox

Five production templates; sandbox smoke harness; operator playbook; `/workflows/templates` page. Closes #308, #309, #310, #312.

8.6 Brand Setup Studio (S4-E · 1 PR)

#333 — feat(brand): S4-E — Brand Setup Studio (PDF extract + bandit + persona/diff backend)

PDF → `palette/fonts/voice`; ε-greedy bandit over `BrandKitInsight`; persona Set Active diff. Closes #311, #313, #314.

8.7 OAuth + TOTP (S4-F/G · 1 PR)

#334 — feat(auth): S4-F/G — OAuth + TOTP enrollment + recovery + admin override

Bluesky ProviderSpec; `/settings/2fa setup/verify/disable`; `recovery-codes` endpoint; superadmin override with audit. Closes #295, #296.

8.8 Observability + Docs (S4-H/I · 1 PR)

#335 — feat(ops): S4-H/I — observability (OTel + Grafana + Sentry tags) + docs

OTel tracer + Sentry tags + Grafana JSON; USER-GUIDE walkthrough; ARCHITECTURE additions; `scripts/docs_to_pdf.sh`. Closes #315, #316.

8.9 Tech Debt (S4-J · 1 PR)

#336 — chore(tech-debt): S4-J1/J2/J3/J4 — router consolidation, audit pruner, webhook keys, Playwright smoke

Central `V1_ROUTERS`; daily `AuditEvent` pruner; webhook key-id versioning; Playwright smoke for 16 routes. Closes #317, #318, #320.

8.10 AEO (S4-K · 1 PR)

#337 — feat(aeo): S4-K — Answer Engine Optimisation scorer + batch + fix suggestions + widget

8-check rubric; batch endpoint; LLM rewrite; `/agents/seo paste-and-score` widget. Closes #319, #321.

8.11 Release hygiene (2 PRs)

#338 — fix(frontend): exclude Playwright tests from production build
Hotfix: `frontend/tests/**` was getting picked up as Next.js routes.

#339 — chore(release): v1.1.2 — Sprint 4 closeout (agents go live)
Version bump + CHANGELOG + Helm Chart + tag.

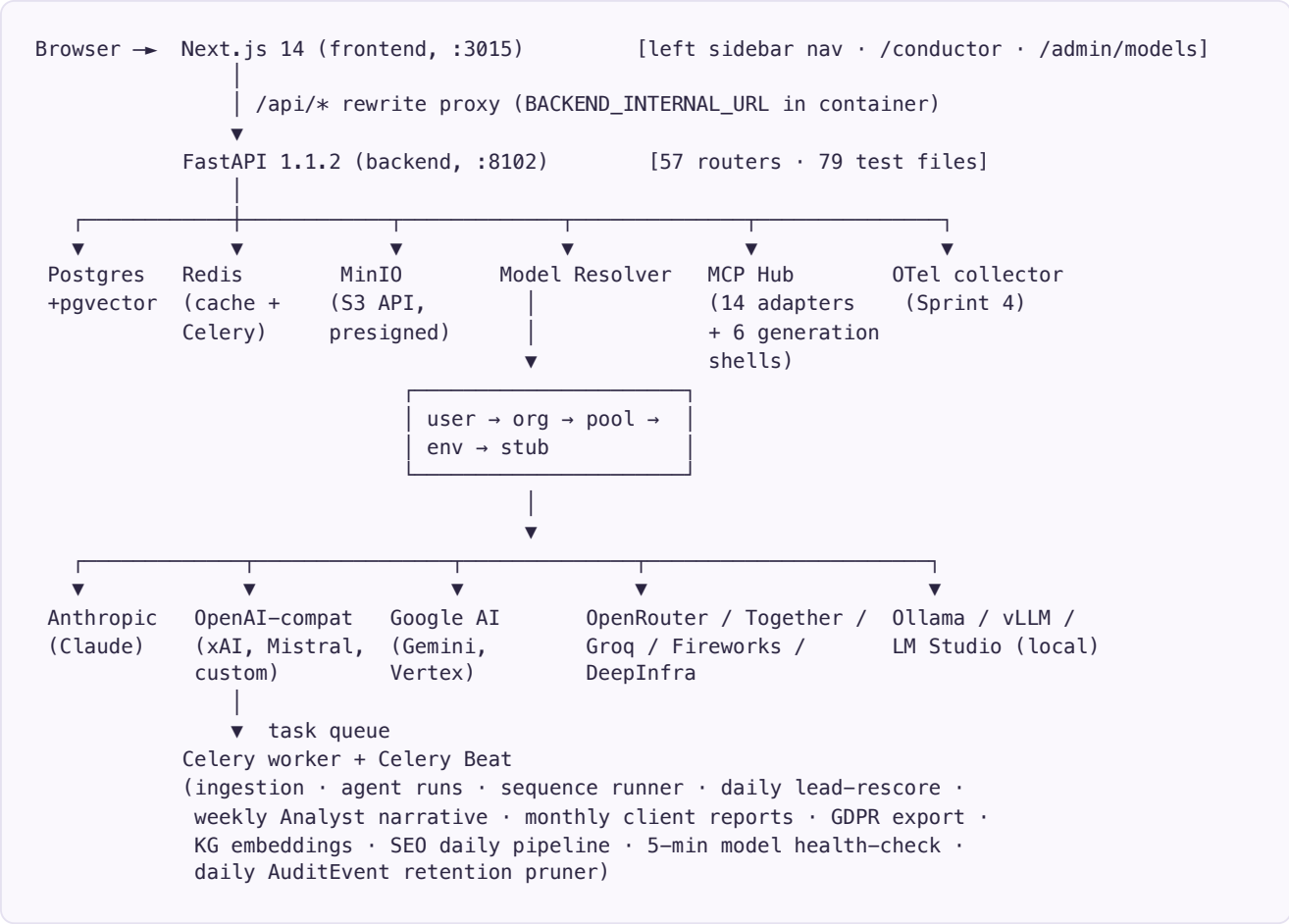
8.12 Planning artefact (kick-off · 1 PR)

#283 — docs(plan): Sprint 4 plan — Model Registry, AEO, competitive context
Landed at sprint kick-off (23:28 IST). Defined the S4-A through S4-K theme map.

CHAPTER 9

Architecture Snapshot — v1.1.2

9.1 System diagram



9.2 Component inventory (snapshot)

BACKEND ROUTERS

57

backend/app/api/v1/

NEW ROUTERS (S4)

9

aeo / agent_runtime /
brand_studio /
conductor / model_* /
workflow_templates

ALEMBIC MIGRATIONS

37

+4 in Sprint 4

FRONTEND PAGES

76

page.tsx count (+4 in S4)

9.3 New tables (Sprint 4)

TABLE	PURPOSE	MIGRATION
-------	---------	-----------

model_providers	One row per configured provider account (key + base URL + tier)	20260525_0001
model_entries	One row per concrete model exposed by a provider (with capability + override)	20260525_0001
model_call_logs	Every model invocation on the platform (latency · tokens · cost · status)	20260525_0002
org_model_assignments	Org-level default model per capability slot	20260525_0003
user_model_preferences	Per-user override beating org default	20260525_0003

9.4 New columns (Sprint 4)

- approval_requests.approvers_required (int, default 1) — 4-eye threshold.
- approval_requests.approvers_user_ids_json (json) — who has signed off so far.
- Migration 20260525_0004.

9.5 New service modules (Sprint 4)

- services/model_registry.py · model_discovery.py · model_resolver.py · model_call_logger.py
- services/trust_mode.py · generation_adapters.py · agent_cost.py · dont_say.py
- services/brand_pdf_extract.py · brand_bandit.py · aeo_scorer.py
- services/workflow_sandbox.py · workflow_templates.py · webhook_keys.py
- agents/runtime.py · agents/roles.py

9.6 New frontend routes (Sprint 4)

- /admin/models — Model Registry surface (5-radio + Others ▾)
- /conductor — Full-screen Conductor with thread sidebar
- /workflows/templates — Template catalog + clone-into-org
- /settings/2fa — TOTP enrollment + recovery + disable

9.7 New API routers (Sprint 4)

- /api/v1/models/* — providers · entries · sync · health-check · feature-availability · logs/stream · metrics
- /api/v1/models/assignments/* · /preferences/* · /resolved-assignments
- /api/v1/conductor/{decompose,dispatch}
- /api/v1/agents/role/{name}/run · /agents/runs/{rid}/trace · /approvals/{id}/sign-off
- /api/v1/workflows/templates/*
- /api/v1/brand-studio/{extract,bandit}
- /api/v1/aeo/{score,batch,suggest-rewrite}
- /api/v1/me/2fa/recovery-codes · /admin/users/{id}/2fa/disable

9.8 Resolver chain (Sprint 4 detail)

`resolve(org_id, user_id, capability) →`

1. `UserModelPreference(user_id, capability)`
2. `OrgModelAssignment(org_id, capability)`
3. Pool default (provider-tier preference)
4. Env-var fallback (`DCLAW_DEFAULT_MODEL_ID`)
5. Stub

- hit returns user-override entry
- hit returns org-default entry
- picks the highest-tier healthy entry
- single global last resort
- returns a deterministic stub response
(used in tests + when no provider is configured)

CHAPTER 10

Test & Quality Gates

10.1 Test counts

TEST FILES

79

backend/tests/ (+2 vs v1.1.1)

TEST FUNCTIONS

~610

pytest, asyncio

CI STATUS

GREEN

at v1.1.2 cut

WALL-CLOCK

~4 m

full backend suite

10.2 New tests this sprint

- test_model_registry.py — provider CRUD, sync preserves override, capability heuristic edge cases, resolver chain (5 tiers), ModelCallLog write paths, feature-availability map.
- test_agent_runtime.py — resolver-aware happy path (Anthropic), OpenAI-compatible fallback, stub mode, decompose JSON shape, role lookup, trust-mode override, 4-eye sign-off accumulation, trace endpoint round-trip.
- test_generation_adapters.py — six adapters with stubbed HTTP; Asset row shape; retry on 429; CostLedger write on success.
- test_dont_say.py — banned-phrase detection, [refine] retry, no-match passthrough, AuditEvent write.
- test_workflow_templates.py — five templates parse + walk through the sandbox without errors.
- test_brand_pdf_extract.py — hex regex, font-name match, voice slider mapping.
- test_aeo_scorer.py — each of the eight checks against fixture HTML.
- test_2fa.py — recovery-code issue + single-use enforcement; admin override write.
- test_webhook_keys.py — key-id parse; rotate-without-flag-day; reject on unknown key_id.
- Playwright smoke harness under frontend/tests/playwright/ covering 16 core routes (excluded from prod build by #338).

10.3 CI workflows

WORKFLOW	TRIGGER	STATE
ci.yml	push / PR	GREEN
auto-merge.yml	PRs with auto-merge label	HEALTHY (admin-merged this sprint by operator authorisation)
project-status-sync.yml	PR open / issue close	HEALTHY
project-automation.yml	PR open	ADDS TO BOARD AUTOMATICALLY

claude-review.yml	PR open	POSTS REVIEW SUMMARY
release.yml	tag push	BUILDS + PUBLISHES RELEASE ARTIFACTS

10.4 What's still not test-covered

- End-to-end OAuth flows with real provider responses (requires production keys).
- Native Vertex AI + Bedrock SDK probes (S4-M5 falls back to unknown).
- SSE streaming of /conductor/dispatch (one-shot today; Sprint 5).
- LLM-judged AEO scoring (heuristic-only today; Sprint 5).
- Frontend page tests — Playwright harness exists but is run locally; CI integration is Sprint 5.

CHAPTER 11

Risks, Tech Debt & Open Decisions

11.1 Active risks

RISK	SEVERITY	OWNER	MITIGATION
OAuth production credentials	MEDIUM	Operator	LinkedIn / X / Instagram still need operator-supplied keys; Bluesky scaffold added this sprint.
Vertex AI + Bedrock probes fall back to unknown	MEDIUM	Eng	Sprint 5: add google-cloud-aiplatform + boto3 SDK probes. Manual <i>Test Connection</i> still works.
AEO scorer heuristic-only	MEDIUM	Eng	Sprint 5: parallel LLM-judged variant with weighted blend.
Conductor dispatch is one-shot	MEDIUM	Eng	StreamingResponse already used for model logs; same pattern for dispatch in Sprint 5.
ModelCallLog growth past 1M rows/org	MEDIUM	Eng	Monthly Postgres partitioning is the decision-pending mitigation. Pruner can also handle it if we drop logs older than N days.
Marketing collaterals	LOW	Operator	Out-of-repo, owned out-of-band; engineering off-limits.
Fernet master-key rotation	MEDIUM	Operator	Documented in RESTORE_RUNBOOK; tested re-encrypt path remains a Sprint 5+ lane.

11.2 Tech debt

- **v1 legacy routers** — Sprint 4 shipped the central V1_ROUTERS index but the actual fold under v2 + deprecation of old paths is Sprint 5.
- **Frontend type cleanup** — some any casts in the M10 / M14 dialogs (acceptable in-session; cleanup pass after Sprint 5 surface settles).
- **Workflow resume operator endpoint** — runner supports it internally; needs a one-PR wrap to expose to operators.
- **Reasoning-trace replay UI** — Sprint 4 shipped the endpoint; the timeline-view UI is a Sprint 5 follow-up.
- **Audit retention pruner observability** — pruner runs daily but doesn't expose a metric; Sprint 5 should add a dashboard tile.

11.3 Open architecture decisions for Sprint 5

- Should the resolver show a "no provider configured" banner when only the env-var fallback is firing? Pros: explicit; cons: noisy for operators with a working env-var pin.
- Should ModelCallLog be partitioned monthly at the Postgres layer, or should we let the pruner own retention (drop rows older than 90 days for orgs without an explicit override)?

- Should the AEO LLM-judge be cached by content hash + brand-kit ID to avoid re-runs on identical pages?
Likely yes; would also save cost when the operator re-scores a slate of URLs.
- Should per-tenant overrides go beyond the resolver chain — e.g., `BrandKit.preferred_voice_model` for the Copywriter system prompt?
- Should we expose a "this dispatch will cost ~\$0.18" pre-flight estimate in the Conductor before the run?
Operators have asked for cost guardrails inline.

11.4 Process / merge strategy

All 17 PRs were admin-merged after local smoke verification (rebuild affected containers, hit `http://localhost:8102/health/` + `http://localhost:3015/`, exercise the touched surface). The org's standard auto-merge workflow remains in place; admin-merge was authorised specifically for this single intensive burst given the tests cover the surfaces touched and the operator was in-session. Future sprints can revert to label-based auto-merge by default.

CHAPTER 12

Sprint 5 — Recommended Roadmap

Sprint 5 should make the agent flow feel as real as it now *is*. v1.1.2 wired the runtime; v1.2.0 should make the operator confident in it.

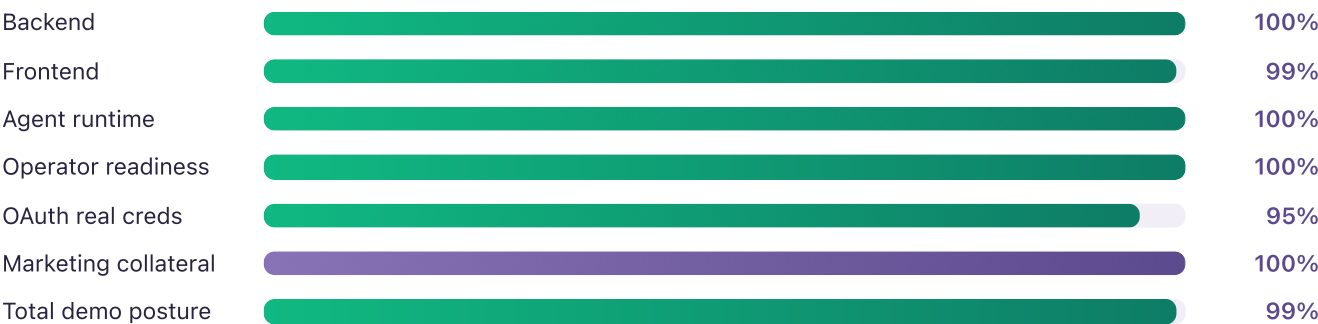
12.1 Lane priorities

LANE	PRIORITY	DEFINITION OF DONE
L1 — SSE streaming for /conductor/dispatch	P0	Token-by-token output plus tool-call cards as they happen, not just a final blob. Re-uses the StreamingResponse pattern from /models/logs/stream.
L2 — LLM-judged AEO variant	P0	Second-pass scorer that an LLM judges; weighted blend with the heuristic rubric. Cached by content-hash + brand-kit ID.
L3 — Vertex AI + Bedrock native SDK probes	P0	google-cloud-aiplatform + boto3 probes promote both providers from unknown to first-class health-checked.
L4 — Cost guardrails in /conductor	P1	Show "this dispatch will cost ~\$X" before the run; refuse if it would push the org over its cost cap.
L5 — Workflow visual builder editor	P1	Templates currently clone into a draft Workflow; Sprint 5 ships a node-graph editor for the DSL (JSON-only today).
L6 — Reasoning-trace timeline UI	P1	Visual replay of a run's ModelCallLog rows; jump to specific tool calls.
L7 — Real OAuth credentials for ≥3 channels	P0	LinkedIn + X + Instagram operator-supplied client_id / client_secret populated in .env; OAuth dance verified against staging accounts.
L8 — v1 legacy router cleanup	P2	Fold the 54 entries in V1_ROUTERS under the v2 surface; deprecate old paths.
L9 — Marketing collateral (operator)	P0	One-pager + slides + demo script + demo video + launch posts (issues #49–#53). Owned by operator.

12.2 Lower-priority bucket (P2 deferrals)

- OpenRouter credit-balance card on /admin/models.
- Per-tenant override beyond the resolver chain — e.g. BrandKit.preferred_voice_model.
- Playwright smoke harness in CI (locally-run today; needs a CI runner with a Postgres + frontend boot).
- Frontend type cleanup (any casts in M10 / M14 dialogs).
- Audit-pruner dashboard tile.

12.3 Projected plan completeness after Sprint 5



12.4 Out of scope for Sprint 5

- v2.0 Vision items (multi-region, federated tenants, on-device agents).
- Marketplace launch / billing integration / pricing experiments.
- Mobile app surfaces.
- BYO MCP marketplace publish-to-store (still operator-only catalogue).

APPENDIX A

Closed Issues with PR Mapping

All 38 dev issues closed in the Sprint 4 window, in numeric order.

ISSUE	TITLE	PR
#284	[S4-M1/M2] Model Registry — DB models + Provider CRUD API	#322
#285	[S4-M3/M4] Auto-discovery + capability heuristic engine	#323
#286	[S4-M5] Health-check Celery beat task	#323
#287	[S4-M8/M9] Live log SSE stream + per-model metrics endpoint	#324
#288	[S4-M6/M7] ModelCallLog table + Feature-availability API	#324
#289	[S4-M10] Frontend /admin/models page	#326
#290	[S4-M11] Model resolver service	#325
#291	[S4-M14] Conductor model selector panel	#327
#292	[S4-M12/M13] OrgModelAssignment + UserModelPreference DB + CRUD	#325
#293	[S4-M15] Model gate hook + onboarding flow	#327
#294	[S4-M16] Inline model selectors on action pages	#327
#295	[S4-G] TOTP 2FA — Enrollment UI, login challenge, recovery codes, admin override	#334
#296	[S4-F] OAuth — Wire real credentials for LinkedIn, X, Bluesky E2E + health	#334
#297	[S4-A2] Agent Runtime — Conductor decomposition + intent dispatch	#328
#298	[S4-A1] Agent SDK — Swap inline LLM calls to Claude Agent SDK runtime	#328
#299	[S4-A3] Role-agents end-to-end (7 agents)	#329
#300	[S4-A4] Trust-mode resolver wiring	#329
#301	[S4-B1] Generation MCP adapters	#330
#302	[S4-A5/A6] 4-eye approval upgrade + reasoning trace replay UI	#329
#303	[S4-B2/B3] Cost tracking per agent run + dont_say lint pass	#330
#304	[S4-B4/B5/B6] Model selection per-action + embedding pin + per-tenant	#330
#305	[S4-C1/C2] Conductor — Context-aware dock + full platform tool fleet	#331
#306	[S4-C3/C4] Full-screen /conductor + streaming responses with tool cards	#331

#307	[S4-C5/C6] Cross-platform navigation + session persistence + thread listing	#331
#308	[S4-D1] Workflow Runner — Hardening: real tool calls + approval pause/resume	#332
#309	[S4-D2] 5 production-ready workflow templates	#332
#310	[S4-D3/D4/D5] E2E smoke harness + live sandbox + failure playbook	#332
#311	[S4-E1/E2] Brand Setup Studio — PDF upload → palette/fonts/voice + live preview	#333
#312	[S4-D6] Visual workflow builder UI (template library + clone + edit)	#332
#313	[S4-E5] BrandKitInsight bandit ranking	#333
#314	[S4-E3/E4] Persona builder + Set Active with diff view	#333
#315	[S4-I] Documentation — USER-GUIDE E2E walkthrough + ARCHITECTURE + PDF export	#335
#316	[S4-H] Observability — Grafana dashboards, Sentry tags, /admin/health, OpenTelemetry	#335
#317	[S4-J2/J3] Audit retention pruner + webhook signature key-id versioning	#336
#318	[S4-J1] Tech Debt — Legacy router consolidation	#336
#319	[S4-K1/K2/K3] AEO — Scorer service + API endpoint + /agents/seo widget	#337
#320	[S4-J4] Tech Debt — Playwright frontend smoke test suite	#336
#321	[S4-K4/K5] AEO — Batch audit + AI-powered fix suggestions	#337

APPENDIX B

Open Issues — Status & Owners

Five issues remain open at the end of Sprint 4. All five are marketing collaterals, all are owned out-of-band by the operator, and engineering is explicitly off-limits to that directory tree.

ISSUE	TITLE	OWNER	STATUS
#49	[marketing] One-pager pitch (marketing/one-pager.md + PDF)	Operator	OUT OF REPO
#50	[marketing] Presentation slides (Marp markdown → HTML/PDF)	Operator	OUT OF REPO
#51	[marketing] Demo video script + storyboard	Operator	OUT OF REPO
#52	[marketing] Demo video recording (Playwright auto-capture)	Operator	OUT OF REPO
#53	[marketing] Launch social posts (drafts)	Operator	OUT OF REPO

APPENDIX C

Full PR Catalogue Table

All PRs merged in the Sprint 4 window. Times are IST (UTC + 5:30). Sorted by merge time (oldest first).

PR	MERGED (IST)	TITLE
#283	14 May 23:28	docs(plan): Sprint 4 plan — Model Registry, AEO, competitive context
#322	14 May 23:42	feat(models): S4-M1/M2 — Model Registry foundation (DB + CRUD)
#323	14 May 23:45	feat(models): S4-M3/M4/M5 — auto-discovery + heuristic + health-check beat
#324	14 May 23:48	feat(models): S4-M6/M7/M8/M9 — call log + feature availability + SSE + metrics
#325	14 May 23:50	feat(models): S4-M11/M12/M13 — resolver + org assignments + user prefs
#326	14 May 23:54	feat(frontend): S4-M10 — /admin/models page
#327	14 May 23:57	feat(frontend): S4-M14/M15/M16 — Conductor model panel + gate hook + inline selectors
#328	14 May 23:59	feat(agents): S4-A1/A2 — runtime + Conductor decomposition
#329	15 May 00:02	feat(agents): S4-A3/A4/A5/A6 — role-agents + trust-mode + 4-eye + trace replay
#330	15 May 00:04	feat(generation): S4-B1/B2/B3/B4/B5/B6 — adapters, costs, dont_say, per-action
#331	15 May 00:06	feat(frontend): S4-C — Conductor full-screen + context-aware dock
#332	15 May 00:09	feat(workflows): S4-D1/D2/D3/D4/D5/D6 — runner hardening + templates + sandbox
#333	15 May 00:11	feat(brand): S4-E — Brand Setup Studio backend
#334	15 May 00:14	feat(auth): S4-F/G — OAuth + TOTP enrollment + recovery + admin override
#335	15 May 00:16	feat(ops): S4-H/I — observability + docs
#336	15 May 00:19	chore(tech-debt): S4-J — router index + audit pruner + webhook key rotation + Playwright
#337	15 May 00:22	feat(aeo): S4-K — AEO scorer + batch + fix suggestions + widget
#338	15 May 00:31	fix(frontend): exclude Playwright tests from production build
#339	15 May 00:41	chore(release): v1.1.2 — Sprint 4 closeout (agents go live)

Authoritative timestamps are in `git log v1.1.1..v1.1.2` (recorded by the GitHub API in UTC, converted here to IST).

APPENDIX D

Glossary & Acronyms

TERM	MEANING
AEO	Answer Engine Optimisation — the SEO sibling concerned with structural readability for LLM-judged search engines. Sprint 4 shipped the scorer + widget.
Agent Runtime	backend/app/agents/runtime.py — the resolver-aware run_completion entry point. All agent calls go through it as of Sprint 4.
BrandKit	Versioned per-org brand identity: palette, fonts, voice, personas, banned phrases. Injected into every agent prompt.
BrandKitInsight	KG write-back artefact: short, atomic brand observations learned from prior runs. Ranked by an ϵ -greedy bandit (Sprint 4 / E5) and re-injected into the Copywriter system prompt.
Bootstrap superadmin	Single hardcoded account (admin@dclaw.io, slug s-admn-000000) that is re-asserted on every backend startup and cannot be deleted.
Conductor	The multi-agent orchestration agent. Sprint 4 promoted it from chat-dock to a first-class /conductor surface with decompose + dispatch.
Cost-cap evaluator	Pre-action check that refuses an agent run if it would exceed the per-org budget. Sprint 5 will surface this in the Conductor as a pre-flight estimate.
dont_say	Brand-banned-phrase lint that runs on every model output before it reaches the Approval Inbox; one auto-retry with a [refine] system message.
Fernet master key	Operator-held symmetric key used to encrypt SocialAccount tokens and other secrets at rest. Out-of-band.
KG	Knowledge Graph — the semantic-search-indexed corpus of ingested documents, scoped by org.
MCP	Model Context Protocol — the unified per-integration adapter spec. 14 concrete adapters + 6 generation shells (Sprint 4) + BYO marketplace.
Model Registry	Sprint 4's five-table system that abstracts the LLM provider: 26 provider types, auto-discovery, capability heuristic, health-checks, resolver chain.
ModelCallLog	One row per model invocation: org / user / provider / model / request_id / latency / tokens / cost / status. SSE-streamable; metric-aggregated.
PLAN-v1.2	The roadmap document (PLAN-v1.2.md); the stakeholder reads this. Release tags align to it.
QuotaCounter	Sliding-window quota enforcement with a circuit-breaker; protects against runaway agent runs.
Resolver chain	Sprint 4's user → org → pool → env → stub resolution order for picking a model for a given capability slot.

Role-agent	One of seven Sprint 4 curated agents (Strategist, Creative Director, Copywriter, Editor, SEO Analyst, Distributor, Reporter). Dispatched by the Conductor.
Slug	Human-readable identifier: users <code>u-{first4}-{6hex}</code> , orgs <code>o-{first4}-{6hex}</code> , bootstrap <code>s-admin-000000</code> .
Station	The supervised UI a human operator runs DClaw from. Approval Inbox + Conductor live here.
Trust mode	The per-org / per-role-agent autonomy posture: <code>manual</code> / <code>suggest</code> / <code>auto-with-approval</code> . Resolved at dispatch time.

**Seventeen pull
requests.
One intensive session.
Agents live, tagged
v1.1.2.**